RFQ Automation Platform - Database Documentation

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Overview

The RFQ Automation Platform database is designed as a multi-tenant SaaS application supporting freight forwarding companies in managing Request for Quotes (RFQs) with shipping carriers. The schema supports complex business workflows including automated email campaigns, AI-powered quote parsing, analytics, and subscription billing.

Key Statistics:

- **Tables:** 23 core tables
- **Primary Technology:** PostgreSQL 12+
- **Multi-tenancy:** Row-level security via company_id
- **Authentication:** JWT-based with refresh tokens
- **Scalability:** Designed for 1000+ companies, millions of RFQs

---

Architecture

System Architecture Layers

```
┌
```

```
│  Application API  │ ← JWT Auth, Rate Limiting
├──────────────────────┤
│ Business Logic    │ ← Multi-tenant isolation
├──────────────────────┤
│ Database Layer    │ ← PostgreSQL with JSONB
├──────────────────────┤
│ Infrastructure    │ ← Redis, Queue System
└──────────────────────┘
```

Multi-Tenant Design Pattern

Tenant Isolation: Row-Level Security (RLS) pattern

- Each tenant (company) has a unique company_id
- All tenant data filtered by company_id
- Global admin data separated in dedicated tables

Core Domain Models

```
Company (Tenant)
├── Users (Admin, Manager, Employee)
├── Shipping Lines (Carriers)
│   └── Contacts
├── RFQs
│   ├── Templates
│   ├── Recipients
│   └── Quotes
├── Email System
│   ├── Templates
│   ├── Logs
│   └── Follow-ups
└── Analytics & Billing
```

Entity Relationship Diagram

Core Entities Flow

```
erDiagram
  COMPANIES ||--o{ COMPANY_USERS : has
  COMPANIES ||--o{ SHIPPING_LINES : manages
  SHIPPING_LINES ||--o{ CONTACTS : contains

  COMPANIES ||--o{ RFQS : creates
  RFQS ||--o{ RFQ_RECIPIENTS : sent_to
  RFQ_RECIPIENTS }o--|| CONTACTS : targets

  RFQS ||--o{ QUOTES : receives
  QUOTES }o--|| CONTACTS : submitted_by
  QUOTES }o--|| SHIPPING_LINES : from

  COMPANIES ||--o{ EMAIL_LOGS : sends
  EMAIL_LOGS }o--o| RFQS : relates_to
  EMAIL_LOGS }o--o| CONTACTS : sent_to
```

Table Specifications

🔐 **Authentication & Identity**

admins

Platform-level administrators with global system access.

Column	Type	Description
id	UUID	Primary key
email	VARCHAR(255)	Unique login email

Column	Type	Description
password_hash	VARCHAR(255)	Bcrypt hashed password
role	VARCHAR(20)	admin or super_admin
is_active	BOOLEAN	Account status

Business Rules:

- Super admins can manage other admins
- Regular admins have read-only global access
- Email must be unique across the platform

companies

Tenant organizations using the platform.

Column	Type	Description
id	UUID	Tenant identifier
name	VARCHAR(255)	Company name
email	VARCHAR(255)	Primary company contact
subscription_plan	VARCHAR(50)	Current billing plan
stripe_customer_id	VARCHAR(100)	Stripe integration
email_footer	TEXT	Custom email signature
is_active	BOOLEAN	Subscription status

Business Rules:

- Each company is completely isolated
- Subscription status controls feature access
- Email domain can be enforced for user registration

company_users

Users within a company tenant.

Column	Type	Description
id	UUID	Primary key
company_id	UUID	Tenant boundary
email	VARCHAR(255)	User login email
role	VARCHAR(20)	admin, manager, employee
is_active	BOOLEAN	Account status

Role Permissions:

- **Admin:** Full company access, user management
- **Manager:** RFQ management, reporting
- **Employee:** RFQ creation, limited access

🚢 Shipping & Contacts

shipping_lines

Carrier companies that provide shipping services.

Column	Type	Description
id	UUID	Primary key
company_id	UUID	Tenant isolation
name	VARCHAR(255)	Carrier name (e.g., "Maersk")
code	VARCHAR(50)	SCAC code or internal ID
is_archived	BOOLEAN	Soft delete flag

Business Rules:

- Name must be unique within company
- Archived carriers hidden from new RFQs
- SCAC codes follow industry standards

contacts

Individual contacts within shipping lines.

Column	Type	Description
id	UUID	Primary key
shipping_line_id	UUID	Parent carrier
email	VARCHAR(255)	Contact email
tags	TEXT[]	Categorization tags
performance_score	INTEGER	AI-calculated score (0-100)
do_not_contact	BOOLEAN	Blacklist flag
is_primary	BOOLEAN	Main contact for carrier

Performance Scoring:

- Based on response rate, quote quality, win rate
- Updated automatically after each RFQ cycle
- Used for contact prioritization

📋 RFQ Management

rfqs

Core request for quote entities.

Column	Type	Description
id	UUID	Primary key
rfq_number	VARCHAR(50)	Human-readable identifier
title	VARCHAR(255)	RFQ description
origin_port	VARCHAR(100)	Shipping origin
destination_port	VARCHAR(100)	Shipping destination
container_type	VARCHAR(50)	20GP, 40GP, 40HC
cargo_ready_date	DATE	When cargo available
status	VARCHAR(20)	Workflow state
best_quote_amount	DECIMAL(12,2)	Lowest received quote

Status Workflow:

draft → sent → (quotes received) → closed/awarded

Business Rules:

- RFQ numbers auto-generated or custom
- Cannot edit sent RFQs
- Status transitions tracked for analytics

rfq_templates

Reusable RFQ configurations.

Column	Type	Description
name	VARCHAR(255)	Template name
subject_template	TEXT	Email subject with tokens
body_template	TEXT	Email body with tokens
origin_port	VARCHAR(100)	Default origin
required_services	TEXT[]	Default service requirements

Token System:

- {{contact_name}} - Personalizes emails
- {{company_name}} - Company branding
- {{rfq_number}} - Reference tracking
- {{cargo_details}} - Shipping specifics

quotes

Responses received from carriers.

Column	Type	Description
id	UUID	Primary key
rfq_id	UUID	Parent RFQ
contact_id	UUID	Submitting contact
total_amount	DECIMAL(12,2)	Final quote price
currency	VARCHAR(3)	ISO currency code

Column	Type	Description
validity_date	DATE	Quote expiration
source	VARCHAR(20)	email, manual, phone
is_parsed_by_ai	BOOLEAN	AI extraction flag
confidence_score	DECIMAL(3,2)	AI confidence (0.00-1.00)
is_winner	BOOLEAN	Awarded quote

AI Parsing:

- Extracts pricing from email text
- Confidence scoring for manual review
- Raw email stored for audit trail

📧 Email System

email_templates

Customizable email templates.

Column	Type	Description
template_type	VARCHAR(50)	rfq, follow_up, award
subject	VARCHAR(500)	Email subject line
body_html	TEXT	Rich HTML content
supported_tokens	TEXT[]	Available placeholders
language	VARCHAR(5)	Localization code

Template Types:

- **RFQ:** Initial quote request
- **Follow-up:** Automated reminders
- **Award:** Winner notification
- **Rejection:** Non-winner notification

email_logs

Complete email activity tracking.

Column	Type	Description
message_id	VARCHAR(255)	Provider tracking ID
status	VARCHAR(20)	Delivery status
to_email	VARCHAR(255)	Recipient address
rfq_id	UUID	Related RFQ (if applicable)
opened_at	TIMESTAMP	Email open time
clicked_at	TIMESTAMP	Link click time

Status Tracking:

queued → sent → delivered → opened → clicked

↓

bounced/failed

scheduled_follow_ups

Automated email follow-up system.

Column	Type	Description
rfq_id	UUID	Target RFQ
contact_id	UUID	Recipient contact
scheduled_at	TIMESTAMP	Send time
follow_up_sequence	INTEGER	1st, 2nd, 3rd follow-up
status	VARCHAR(20)	Execution status

Follow-up Logic:

- Triggered by configurable rules
- Skipped if contact has responded
- Maximum sequence limit per company

📊 Analytics & Reporting

carrier_performance_metrics

Aggregated performance data per carrier.

Column	Type	Description
shipping_line_id	UUID	Target carrier
period_start/end	DATE	Measurement period
response_rate	DECIMAL(5,2)	Percentage responded
average_response_time_hours	DECIMAL(8,2)	Speed metric
win_rate	DECIMAL(5,2)	Awarded quotes %
average_quote_amount	DECIMAL(12,2)	Pricing average

Calculation Schedule:

- Daily: Real-time updates
- Monthly: Materialized aggregations
- Quarterly: Trend analysis

usage_metrics

Subscription billing and usage tracking.

Column	Type	Description
company_id	UUID	Tenant identifier
period_month/year	INTEGER	Billing period
rfqs_created	INTEGER	Monthly RFQ count
emails_sent	INTEGER	Email volume
ai_parsing_requests	INTEGER	AI service usage

Usage Limits:

- Enforced at API level
- Alerts at 80% of limit
- Automatic overage billing

🔌 Integrations

webhook_endpoints

External system notifications.

Column	Type	Description
url	VARCHAR(500)	Target endpoint
events	TEXT[]	Subscribed event types
secret	VARCHAR(255)	HMAC signature key
failure_count	INTEGER	Health monitoring

Event Types:

- quote.received - New quote parsed
- rfq.sent - RFQ distributed
- email.opened - Engagement tracking
- quote.awarded - Winner selected

api_keys

External API access credentials.

Column	Type	Description
api_key	VARCHAR(255)	Hashed access token
scopes	TEXT[]	Permission boundaries
rate_limit_per_hour	INTEGER	Usage throttling
expires_at	TIMESTAMP	Key expiration

Scope System:

- rfq:read - View RFQs
- rfq:write - Create/modify RFQs
- quotes:read - Access quote data
- webhook:manage - Configure integrations

Multi-Tenancy Implementation

Data Isolation Strategy

Row-Level Security (RLS)

-- Example policy for RFQs table

```
CREATE POLICY company_rfqs_policy ON rfqs
  FOR ALL TO application_role
  USING (company_id = current_setting('app.current_company_id')::UUID);
```

Application-Level Filtering

```
-- Every tenant query must include company_id
SELECT * FROM rfqs
WHERE company_id = $1 AND status = 'sent';
```

Tenant Context Management

Session Variables:

- app.current_company_id - Set on connection
- app.current_user_id - User context
- app.user_role - Permission level

Connection Pooling:

- Dedicated pools per tenant (high-traffic)
- Shared pool with session reset (cost-effective)

Data Flow & Business Logic

RFQ Lifecycle

```
graph TD
  A[Create Draft RFQ] --> B[Select Contacts]
  B --> C[Send Email Batch]
  C --> D[Track Delivery]
  D --> E[Receive Quotes]
  E --> F[AI Parse Quotes]
  F --> G[Compare & Analyze]
  G --> H[Award Winner]
  H --> I[Send Notifications]
```

Email Campaign Flow

1. **Template Selection:** Choose RFQ template
2. **Contact Filtering:** Apply tags, performance scores
3. **Personalization:** Replace tokens with data
4. **Batch Processing:** Queue management for rate limiting
5. **Delivery Tracking:** Webhook integration with providers
6. **Follow-up Automation:** Rule-based scheduling

Quote Processing Pipeline

1. **Email Ingestion:** IMAP or webhook receive
2. **RFQ Matching:** Thread/subject parsing
3. **AI Extraction:** Price, terms, validity parsing
4. **Confidence Scoring:** Manual review flagging
5. **Data Validation:** Business rule enforcement
6. **Notification:** Real-time updates to users

Indexing Strategy

Multi-Tenant Indexes

Composite Indexes for Tenant Queries:

```
CREATE INDEX idx_rfqs_company_status ON rfqs(company_id, status);

CREATE INDEX idx_quotes_rfq_status ON quotes(rfq_id, status);

CREATE INDEX idx_contacts_company_active ON contacts(company_id, is_active);
```

Performance-Critical Indexes

Email System:

```
CREATE INDEX idx_email_logs_tracking ON email_logs(message_id, status);

CREATE INDEX idx_scheduled_followups_execution

ON scheduled_follow_ups(scheduled_at, status) WHERE status = 'scheduled';
```

Analytics Queries:

```
CREATE INDEX idx_quotes_analytics

ON quotes(rfq_id, created_at, total_amount) WHERE status = 'active';
```

Partial Indexes

Active Records Only:

```
CREATE INDEX idx_contacts_active
ON contacts(company_id, shipping_line_id) WHERE is_active = true;
```

Security Considerations

Data Protection

Encryption:

- Passwords: bcrypt with salt rounds 12+
- API keys: SHA-256 hashed storage
- Sensitive data: Application-level encryption for PII

Access Control:

- JWT tokens with short expiration (15 min)
- Refresh token rotation
- Role-based permissions at API level

Audit Trail

Change Tracking:

```
-- Trigger-based audit logging
CREATE TABLE audit_log (
  table_name VARCHAR(50),
  operation VARCHAR(10),
  old_data JSONB,
  new_data JSONB,
  changed_by UUID,
  changed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Critical Events:

- RFQ status changes

- Quote awards
- User permission modifications
- Billing plan changes

Data Retention

Automatic Cleanup:

- Email logs: 2 years
- Audit trails: 7 years
- Expired quotes: 1 year after RFQ closure
- Inactive companies: Anonymize after 3 years

Performance Guidelines

Query Optimization

Always Include Tenant Filter:

```
-- ✅ Good
SELECT * FROM rfqs WHERE company_id = $1 AND status = 'sent';


-- ❌ Bad - Missing tenant boundary
SELECT * FROM rfqs WHERE status = 'sent';
```

Use Prepared Statements:

```
PREPARE get_company_rfqs AS
SELECT id, title, status FROM rfqs
WHERE company_id = $1 AND created_at >= $2;
```

Connection Management

Pool Configuration:

- Min connections: 5
- Max connections: 100
- Idle timeout: 300 seconds
- Statement timeout: 30 seconds

Monitoring Queries

Slow Query Detection:

```
-- Enable pg_stat_statements
SELECT query, calls, total_time, mean_time
FROM pg_stat_statements
WHERE mean_time > 1000
ORDER BY mean_time DESC;
```

Migration & Deployment

Schema Versioning

Migration Structure:

```
migrations/
├── 001_initial_schema.sql
├── 002_add_ai_parsing.sql
├── 003_webhook_system.sql
└── rollback/
    ├── 001_rollback.sql
    └── 002_rollback.sql
```

Deployment Process

1. **Pre-deployment:**
    - Backup production database
    - Run migration in staging
    - Performance test new queries
2. **Migration Execution:**
    - Run during low-traffic window
    - Monitor connection count
    - Validate data integrity
3. **Post-deployment:**

    - Update application configuration
    - Monitor error rates
    - Verify key functionality

Zero-Downtime Considerations

Additive Changes:

- New columns with defaults
- New indexes (CREATE INDEX CONCURRENTLY)
- New tables

Breaking Changes:

- Column renaming: Use views during transition
- Data type changes: Multi-phase deployment
- Constraint additions: Validate existing data first

Maintenance & Monitoring

Regular Maintenance Tasks

Weekly:

- VACUUM ANALYZE on high-churn tables
- Index usage analysis
- Connection pool monitoring

Monthly:

- Update table statistics
- Analyze slow query log
- Archive old audit data

Quarterly:

- Full database backup verification
- Capacity planning review
- Performance benchmark comparison

Health Metrics

Database Health:

- Connection utilization < 80%
- Average query time < 100ms
- Lock wait time < 5ms
- Cache hit ratio > 95%

Business Metrics:

- Email delivery rate > 95%
- AI parsing accuracy > 90%
- Average RFQ response time
- Quote processing latency

Conclusion

This database schema provides a robust foundation for the RFQ Automation Platform, supporting:

- **Scalability:** Multi-tenant architecture for thousands of companies
- **Performance:** Optimized indexes and query patterns
- **Flexibility:** JSONB fields for evolving requirements
- **Reliability:** Comprehensive audit trails and data integrity
- **Integration:** API-first design with webhook support

The schema balances normalization with practical performance needs, ensuring both data consistency and query efficiency as the platform scales.

```
-- RFQ Automation Platform - Complete Database Schema
-- PostgreSQL with Multi-Tenant Architecture


-- Enable UUID extension
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";


-- ==============================================
```

```
-- CORE USER & AUTHENTICATION TABLES
-- ==============================================

-- Global Admins (Platform Level)
CREATE TABLE admins (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    email VARCHAR(255) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100) NOT NULL,
    role VARCHAR(20) NOT NULL DEFAULT 'admin' CHECK (role IN ('admin',
'super_admin')),
    is_active BOOLEAN DEFAULT true,
    last_login_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Companies (Tenant Level)
CREATE TABLE companies (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    name VARCHAR(255) NOT NULL,
    email VARCHAR(255) UNIQUE NOT NULL,
    domain VARCHAR(100), -- For email domain validation
    phone VARCHAR(50),
    address TEXT,
    city VARCHAR(100),
    country VARCHAR(100),
```

```
    timezone VARCHAR(50) DEFAULT 'UTC',

    -- Subscription & Billing
    subscription_plan VARCHAR(50) DEFAULT 'trial',
    subscription_status VARCHAR(20) DEFAULT 'active' CHECK (subscription_status IN
('active', 'inactive', 'suspended', 'canceled')),
    stripe_customer_id VARCHAR(100),
    trial_ends_at TIMESTAMP,

    -- Company Settings
    email_footer TEXT,
    default_follow_up_days INTEGER DEFAULT 3,
    auto_follow_up_enabled BOOLEAN DEFAULT true,

    -- Status
    is_active BOOLEAN DEFAULT true,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Company Users (Tenant Users)
CREATE TABLE company_users (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
    email VARCHAR(255) NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100) NOT NULL,
```

```
    role VARCHAR(20) NOT NULL DEFAULT 'employee' CHECK (role IN ('admin',
'manager', 'employee')),
    is_active BOOLEAN DEFAULT true,
    last_login_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(company_id, email)
);

-- JWT Refresh Tokens
CREATE TABLE refresh_tokens (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    token_hash VARCHAR(255) NOT NULL,
    user_id UUID, -- Can be admin or company_user
    user_type VARCHAR(20) NOT NULL CHECK (user_type IN ('admin',
'company_user')),
    company_id UUID REFERENCES companies(id) ON DELETE CASCADE,
    expires_at TIMESTAMP NOT NULL,
    is_revoked BOOLEAN DEFAULT false,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- ==============================================
-- SHIPPING LINES & CONTACTS
-- ==============================================

-- Shipping Lines (Carriers)
CREATE TABLE shipping_lines (
```

```
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
    name VARCHAR(255) NOT NULL,
    code VARCHAR(50), -- SCAC code or internal code
    website VARCHAR(255),
    headquarters_location VARCHAR(100),
    description TEXT,
    is_active BOOLEAN DEFAULT true,
    is_archived BOOLEAN DEFAULT false,
    created_by UUID NOT NULL REFERENCES company_users(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(company_id, name)
);

-- Contacts for Shipping Lines
CREATE TABLE contacts (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
    shipping_line_id UUID NOT NULL REFERENCES shipping_lines(id) ON DELETE
CASCADE,

    -- Contact Info
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100) NOT NULL,
    email VARCHAR(255) NOT NULL,
    phone VARCHAR(50),
```

```
    job_title VARCHAR(100),
    department VARCHAR(100),

    -- Contact Management
    tags TEXT[], -- PostgreSQL array for tags
    notes TEXT,
    performance_score INTEGER DEFAULT 0 CHECK (performance_score >= 0 AND
performance_score <= 100),
    do_not_contact BOOLEAN DEFAULT false,
    is_primary BOOLEAN DEFAULT false,

    -- Status
    is_active BOOLEAN DEFAULT true,
    is_archived BOOLEAN DEFAULT false,
    created_by UUID NOT NULL REFERENCES company_users(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(company_id, shipping_line_id, email)
);

-- ==============================================
-- RFQ MANAGEMENT
-- ==============================================

-- RFQ Templates
CREATE TABLE rfq_templates (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
```

```
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
    name VARCHAR(255) NOT NULL,
    description TEXT,

    -- RFQ Content
    subject_template TEXT NOT NULL,
    body_template TEXT NOT NULL,

    -- Shipping Details Template
    origin_port VARCHAR(100),
    destination_port VARCHAR(100),
    commodity VARCHAR(255),
    container_type VARCHAR(50),
    container_quantity INTEGER,
    cargo_weight DECIMAL(10,2),
    cargo_volume DECIMAL(10,2),
    incoterm VARCHAR(10),

    -- Additional Requirements
    special_requirements TEXT,
    required_services TEXT[], -- ['customs_clearance', 'insurance', 'door_delivery']

    is_default BOOLEAN DEFAULT false,
    created_by UUID NOT NULL REFERENCES company_users(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

```
-- RFQs (Request for Quotes)
CREATE TABLE rfqs (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
    rfq_number VARCHAR(50) NOT NULL, -- Auto-generated or custom

    -- RFQ Details
    title VARCHAR(255) NOT NULL,
    description TEXT,

    -- Shipping Details
    origin_port VARCHAR(100) NOT NULL,
    destination_port VARCHAR(100) NOT NULL,
    commodity VARCHAR(255),
    container_type VARCHAR(50), -- '20GP', '40GP', '40HC', etc.
    container_quantity INTEGER DEFAULT 1,
    cargo_weight DECIMAL(10,2),
    cargo_volume DECIMAL(10,2),
    incoterm VARCHAR(10) DEFAULT 'FOB',

    -- Timeline
    cargo_ready_date DATE,
    quote_deadline TIMESTAMP,
    shipment_urgency VARCHAR(20) DEFAULT 'normal' CHECK (shipment_urgency IN
('urgent', 'normal', 'flexible')),

    -- Additional Requirements
    special_requirements TEXT,
```

```
    required_services TEXT[], -- JSON array of required services
    preferred_carriers UUID[], -- Array of shipping_line IDs

    -- Email Content
    email_subject TEXT,
    email_body TEXT,

    -- Status & Workflow
    status VARCHAR(20) NOT NULL DEFAULT 'draft' CHECK (status IN ('draft', 'sent',
'closed', 'awarded', 'expired')),
    sent_at TIMESTAMP,
    closed_at TIMESTAMP,

    -- Metrics
    total_contacts_sent INTEGER DEFAULT 0,
    total_quotes_received INTEGER DEFAULT 0,
    best_quote_amount DECIMAL(12,2),
    average_quote_amount DECIMAL(12,2),

    -- Audit
    created_by UUID NOT NULL REFERENCES company_users(id),
    assigned_to UUID REFERENCES company_users(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(company_id, rfq_number)
);
```

```
-- RFQ Recipients (Contacts who received the RFQ)
CREATE TABLE rfq_recipients (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    rfq_id UUID NOT NULL REFERENCES rfqs(id) ON DELETE CASCADE,
    contact_id UUID NOT NULL REFERENCES contacts(id) ON DELETE CASCADE,
    shipping_line_id UUID NOT NULL REFERENCES shipping_lines(id) ON DELETE
CASCADE,

    -- Email Tracking
    email_sent_at TIMESTAMP NOT NULL,
    email_delivered_at TIMESTAMP,
    email_opened_at TIMESTAMP,
    email_clicked_at TIMESTAMP,
    last_email_activity_at TIMESTAMP,

    -- Response Tracking
    has_responded BOOLEAN DEFAULT false,
    response_received_at TIMESTAMP,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(rfq_id, contact_id)
);

-- ==============================================
-- QUOTE MANAGEMENT
-- ==============================================
```

```
-- Quotes received for RFQs
CREATE TABLE quotes (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    rfq_id UUID NOT NULL REFERENCES rfqs(id) ON DELETE CASCADE,
    contact_id UUID NOT NULL REFERENCES contacts(id),
    shipping_line_id UUID NOT NULL REFERENCES shipping_lines(id),

    -- Quote Details
    quote_reference VARCHAR(100), -- Carrier's internal quote reference

    -- Pricing
    ocean_freight DECIMAL(12,2),
    currency VARCHAR(3) DEFAULT 'USD',

    -- Additional Charges
    baf DECIMAL(12,2), -- Bunker Adjustment Factor
    caf DECIMAL(12,2), -- Currency Adjustment Factor
    security_fee DECIMAL(12,2),
    documentation_fee DECIMAL(12,2),
    handling_charges DECIMAL(12,2),
    other_charges JSONB, -- Flexible structure for other charges

    -- Total
    total_amount DECIMAL(12,2) NOT NULL,

    -- Terms & Conditions
    validity_date DATE NOT NULL,
    payment_terms VARCHAR(100),
```

```
    transit_time VARCHAR(50), -- "15-20 days"
    free_time_at_origin INTEGER, -- days
    free_time_at_destination INTEGER, -- days

    -- Additional Terms
    terms_and_conditions TEXT,
    special_notes TEXT,

    -- Source & Processing
    source VARCHAR(20) DEFAULT 'email' CHECK (source IN ('email', 'manual', 'phone',
'whatsapp')),
    is_parsed_by_ai BOOLEAN DEFAULT false,
    confidence_score DECIMAL(3,2), -- AI parsing confidence (0.00-1.00)
    raw_email_content TEXT, -- Store original email for reference

    -- Status
    status VARCHAR(20) DEFAULT 'active' CHECK (status IN ('active', 'expired',
'withdrawn', 'awarded')),
    is_winner BOOLEAN DEFAULT false,

    -- Audit
    created_by UUID REFERENCES company_users(id), -- NULL if AI parsed
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- ==============================================
-- EMAIL SYSTEM
-- ==============================================
```

```
-- Email Templates
CREATE TABLE email_templates (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
    name VARCHAR(255) NOT NULL,
    template_type VARCHAR(50) NOT NULL, -- 'rfq', 'follow_up', 'award', 'rejection'

    -- Template Content
    subject VARCHAR(500) NOT NULL,
    body_html TEXT NOT NULL,
    body_text TEXT,

    -- Personalization
    supported_tokens TEXT[], -- ['{{contact_name}}', '{{company_name}}', etc.]

    -- Multi-language Support
    language VARCHAR(5) DEFAULT 'en',

    is_default BOOLEAN DEFAULT false,
    is_active BOOLEAN DEFAULT true,

    created_by UUID NOT NULL REFERENCES company_users(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(company_id, name)
);
```

```
-- Email Send Log
CREATE TABLE email_logs (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,

    -- Email Details
    to_email VARCHAR(255) NOT NULL,
    from_email VARCHAR(255) NOT NULL,
    subject VARCHAR(500) NOT NULL,
    body_html TEXT,
    body_text TEXT,

    -- Tracking
    message_id VARCHAR(255), -- Provider message ID
    status VARCHAR(20) DEFAULT 'queued' CHECK (status IN ('queued', 'sent',
'delivered', 'opened', 'clicked', 'bounced', 'failed')),

    -- Relationships
    rfq_id UUID REFERENCES rfqs(id),
    contact_id UUID REFERENCES contacts(id),
    template_id UUID REFERENCES email_templates(id),

    -- Timestamps
    queued_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    sent_at TIMESTAMP,
    delivered_at TIMESTAMP,
    opened_at TIMESTAMP,
```

```
    clicked_at TIMESTAMP,
    bounced_at TIMESTAMP,
    failed_at TIMESTAMP,

    -- Error Handling
    error_message TEXT,
    retry_count INTEGER DEFAULT 0,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- ==============================================
-- AUTOMATED FOLLOW-UPS
-- ==============================================

-- Follow-up Rules
CREATE TABLE follow_up_rules (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,

    name VARCHAR(255) NOT NULL,
    description TEXT,

    -- Trigger Conditions
    days_after_send INTEGER NOT NULL DEFAULT 3,
    only_if_not_opened BOOLEAN DEFAULT false,
    only_if_not_replied BOOLEAN DEFAULT true,
    max_follow_ups INTEGER DEFAULT 2,
```

```
    -- Email Template
    email_template_id UUID REFERENCES email_templates(id),

    is_active BOOLEAN DEFAULT true,
    created_by UUID NOT NULL REFERENCES company_users(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Scheduled Follow-ups
CREATE TABLE scheduled_follow_ups (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    rfq_id UUID NOT NULL REFERENCES rfqs(id) ON DELETE CASCADE,
    contact_id UUID NOT NULL REFERENCES contacts(id) ON DELETE CASCADE,
    follow_up_rule_id UUID NOT NULL REFERENCES follow_up_rules(id),

    -- Scheduling
    scheduled_at TIMESTAMP NOT NULL,
    sent_at TIMESTAMP,

    -- Status
    status VARCHAR(20) DEFAULT 'scheduled' CHECK (status IN ('scheduled', 'sent',
'skipped', 'failed')),
    skip_reason VARCHAR(100), -- 'already_replied', 'contact_inactive', etc.

    -- Sequence
    follow_up_sequence INTEGER DEFAULT 1, -- 1st, 2nd, 3rd follow-up
```

```
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- ==============================================
-- ANALYTICS & REPORTING
-- ==============================================

-- Carrier Performance Metrics (Materialized for performance)
CREATE TABLE carrier_performance_metrics (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,
    shipping_line_id UUID NOT NULL REFERENCES shipping_lines(id) ON DELETE
CASCADE,

    -- Time Period
    period_start DATE NOT NULL,
    period_end DATE NOT NULL,
    period_type VARCHAR(20) NOT NULL, -- 'monthly', 'quarterly', 'yearly'

    -- Metrics
    rfqs_received INTEGER DEFAULT 0,
    quotes_submitted INTEGER DEFAULT 0,
    response_rate DECIMAL(5,2) DEFAULT 0, -- Percentage
    average_response_time_hours DECIMAL(8,2),
    quotes_won INTEGER DEFAULT 0,
    win_rate DECIMAL(5,2) DEFAULT 0,
    average_quote_amount DECIMAL(12,2),
```

```
    -- Calculated at end of period
    calculated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(company_id, shipping_line_id, period_start, period_end, period_type)
);

-- Route Performance Metrics
CREATE TABLE route_performance_metrics (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,

    -- Route Definition
    origin_port VARCHAR(100) NOT NULL,
    destination_port VARCHAR(100) NOT NULL,

    -- Time Period
    period_start DATE NOT NULL,
    period_end DATE NOT NULL,
    period_type VARCHAR(20) NOT NULL,

    -- Metrics
    total_rfqs INTEGER DEFAULT 0,
    total_quotes INTEGER DEFAULT 0,
    average_quote_amount DECIMAL(12,2),
    lowest_quote_amount DECIMAL(12,2),
    highest_quote_amount DECIMAL(12,2),
    most_competitive_carrier_id UUID REFERENCES shipping_lines(id),
```

```
    calculated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(company_id, origin_port, destination_port, period_start, period_end,
period_type)
);

-- ==============================================
-- SYSTEM FEATURES & BILLING
-- ==============================================

-- System Features (for subscription management)
CREATE TABLE system_features (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    name VARCHAR(100) NOT NULL UNIQUE,
    description TEXT,
    feature_key VARCHAR(50) NOT NULL UNIQUE, -- 'max_rfqs_per_month',
'email_tracking', etc.
    feature_type VARCHAR(20) NOT NULL CHECK (feature_type IN ('boolean', 'limit',
'addon')),
    is_active BOOLEAN DEFAULT true,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Subscription Plans
CREATE TABLE subscription_plans (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    name VARCHAR(100) NOT NULL,
    description TEXT,
```

```
    price_monthly DECIMAL(10,2),
    price_yearly DECIMAL(10,2),

    -- Limits
    max_users INTEGER,
    max_rfqs_per_month INTEGER,
    max_contacts INTEGER,
    max_email_sends_per_month INTEGER,

    -- Features
    features JSONB, -- {'email_tracking': true, 'ai_parsing': true, 'analytics': 'advanced'}

    is_active BOOLEAN DEFAULT true,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Usage Tracking
CREATE TABLE usage_metrics (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,

    -- Period
    period_month INTEGER NOT NULL,
    period_year INTEGER NOT NULL,

    -- Usage Counts
    rfqs_created INTEGER DEFAULT 0,
    emails_sent INTEGER DEFAULT 0,
```

```
    ai_parsing_requests INTEGER DEFAULT 0,
    api_calls INTEGER DEFAULT 0,

    -- Storage Usage (in MB)
    file_storage_mb DECIMAL(10,2) DEFAULT 0,

    last_updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(company_id, period_month, period_year)
);

-- ==============================================
-- INTEGRATIONS & API
-- ==============================================

-- Webhook Endpoints
CREATE TABLE webhook_endpoints (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,

    name VARCHAR(255) NOT NULL,
    url VARCHAR(500) NOT NULL,
    secret VARCHAR(255), -- For HMAC signature verification

    -- Event Subscriptions
    events TEXT[] NOT NULL, -- ['quote.received', 'rfq.sent', 'email.opened']

    -- Status & Health
```

```
    is_active BOOLEAN DEFAULT true,
    last_success_at TIMESTAMP,
    last_failure_at TIMESTAMP,
    failure_count INTEGER DEFAULT 0,

    created_by UUID NOT NULL REFERENCES company_users(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- API Keys (for external integrations)
CREATE TABLE api_keys (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID NOT NULL REFERENCES companies(id) ON DELETE CASCADE,

    key_name VARCHAR(255) NOT NULL,
    api_key VARCHAR(255) UNIQUE NOT NULL, -- Hashed

    -- Permissions
    scopes TEXT[] NOT NULL, -- ['rfq:read', 'rfq:write', 'quotes:read']

    -- Rate Limiting
    rate_limit_per_hour INTEGER DEFAULT 1000,

    -- Status
    is_active BOOLEAN DEFAULT true,
    last_used_at TIMESTAMP,
    expires_at TIMESTAMP,
```

```
    created_by UUID NOT NULL REFERENCES company_users(id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- ==============================================
-- SUPPORT SYSTEM
-- ==============================================

-- Support Tickets
CREATE TABLE support_tickets (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    company_id UUID REFERENCES companies(id) ON DELETE SET NULL,
    user_id UUID, -- Can be admin or company user
    user_type VARCHAR(20) CHECK (user_type IN ('admin', 'company_user')),

    -- Ticket Details
    title VARCHAR(255) NOT NULL,
    description TEXT NOT NULL,
    priority VARCHAR(20) DEFAULT 'normal' CHECK (priority IN ('low', 'normal', 'high',
'urgent')),
    category VARCHAR(50), -- 'bug', 'feature_request', 'billing', 'general'

    -- Status
    status VARCHAR(20) DEFAULT 'open' CHECK (status IN ('open', 'in_progress',
'resolved', 'closed')),
    assigned_to UUID REFERENCES admins(id),

    -- Timestamps
```

```
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    resolved_at TIMESTAMP,
    closed_at TIMESTAMP
);

-- ==============================================
-- INDEXES FOR PERFORMANCE
-- ==============================================

-- Authentication & Users
CREATE INDEX idx_company_users_email ON company_users(email);
CREATE INDEX idx_company_users_company ON company_users(company_id);
CREATE INDEX idx_refresh_tokens_user ON refresh_tokens(user_id, user_type);

-- Multi-tenant indexes
CREATE INDEX idx_shipping_lines_company ON shipping_lines(company_id);
CREATE INDEX idx_contacts_company ON contacts(company_id);
CREATE INDEX idx_contacts_shipping_line ON contacts(shipping_line_id);
CREATE INDEX idx_rfqs_company ON rfqs(company_id);
CREATE INDEX idx_rfqs_status ON rfqs(status);
CREATE INDEX idx_quotes_rfq ON quotes(rfq_id);

-- Email system indexes
CREATE INDEX idx_email_logs_company ON email_logs(company_id);
CREATE INDEX idx_email_logs_status ON email_logs(status);
CREATE INDEX idx_email_logs_rfq ON email_logs(rfq_id);

-- Performance metrics indexes
```

```
CREATE INDEX idx_carrier_metrics_company ON
carrier_performance_metrics(company_id);

CREATE INDEX idx_carrier_metrics_period ON
carrier_performance_metrics(period_start, period_end);

CREATE INDEX idx_route_metrics_company ON
route_performance_metrics(company_id);


-- ==============================================

-- TRIGGERS FOR UPDATED_AT

-- ==============================================


-- Function to update updated_at timestamp

CREATE OR REPLACE FUNCTION update_updated_at_column()

RETURNS TRIGGER AS $$

BEGIN

   NEW.updated_at = CURRENT_TIMESTAMP;

   RETURN NEW;

END;

$$ language 'plpgsql';


-- Apply to relevant tables

CREATE TRIGGER update_companies_updated_at BEFORE UPDATE ON companies
FOR EACH ROW EXECUTE FUNCTION update_updated_at_column();

CREATE TRIGGER update_company_users_updated_at BEFORE UPDATE ON
company_users FOR EACH ROW EXECUTE FUNCTION update_updated_at_column();

CREATE TRIGGER update_shipping_lines_updated_at BEFORE UPDATE ON
shipping_lines FOR EACH ROW EXECUTE FUNCTION update_updated_at_column();

CREATE TRIGGER update_contacts_updated_at BEFORE UPDATE ON contacts FOR
EACH ROW EXECUTE FUNCTION update_updated_at_column();
```

```
CREATE TRIGGER update_rfqs_updated_at BEFORE UPDATE ON rfqs FOR EACH ROW EXECUTE FUNCTION update_updated_at_column();

CREATE TRIGGER update_quotes_updated_at BEFORE UPDATE ON quotes FOR EACH ROW EXECUTE FUNCTION update_updated_at_column();
```